Notes: Cookson, Niessner, and Schiller (JF, 2026)

Can Social Media Inform Corporate Decisions? Evidence from Merger Withdrawals

Contents

1 Summary

Cookson, Niessner, and Schiller (2026) ask whether *investor social-media sentiment* contains information that is used in *corporate decisions*. They focus on mergers and acquisitions (M&A), where (i) the announcement date is sharp and (ii) the outcome is observable (deal completion versus withdrawal). Using StockTwits (an investor-focused social platform with ticker-tagged posts and sentiment labels), they construct a deal-specific measure of *abnormal social-media sentiment* around the announcement. Their core empirical result is that more negative abnormal sentiment predicts a higher probability that a deal is later withdrawn, even after controlling for traditional market reactions, news sentiment, and analyst recommendation changes. They provide mechanism evidence consistent with *manager learning from social media*: the informativeness of sentiment strengthens after firms create corporate Twitter accounts (especially active/verified/high-follower accounts), and the signal is driven by longer, fundamentally oriented tweets (and by fundamental topics such as deal terms), not by memes or technical-trading content. Finally, the market appears to *welcome* the withdrawal of deals that social media disliked at announcement, consistent with social media identifying value-destroying deals.

2 Empirical question, hypothesis, and why mergers are a clean setting

2.1 Main question

Does social media provide *useful* information for firm managers when making corporate decisions?

2.2 Core hypothesis (manager learning channel)

If announced deals reflect genuine managerial intent to proceed, then later withdrawal indicates an *update* in managerial beliefs about the desirability/feasibility of the acquisition. If the *initial* social-media reaction to the announcement predicts eventual withdrawal, holding constant other information sources that affect withdrawal, then social media sentiment is informative for corporate decision-making.

2.3 Why M&A withdrawals are informative

View the "withdrawal" outcome as attractive empirically because:

- **High-stakes decision:** A merger is among the largest investments a firm makes.
- **Sharp event timing:** Announcement dates are precisely identified, which is ideal for measuring social-media "reaction windows".
- **Observable state change:** Completion vs. withdrawal is a clear discrete outcome that can be modeled with limited assumptions.
- **Interpretability:** Holding other signals constant, withdrawal is consistent with a managerial update (information arrives between announcement and conclusion).

3 Data and measurement

3.1 M&A sample

The paper studies U.S. acquirers and announced acquisitions from 2010–2021, with a minimum deal value threshold (reported as $25M). Deal characteristics are from SDC Platinum, accounting variables from Compustat, and stock returns from CRSP. News sentiment/coverage are from RavenPack, and analyst recommendation variables are included (net changes and counts).

3.2 StockTwits data and deal-level social-media sentiment

StockTwits is an investor social platform where users post short messages ("tweets") and use *cashtags* (e.g., $AAPL) to reference specific firms. StockTwits provides a sentiment label (bullish/bearish), allowing aggregation to a stock-day sentiment measure.

3.3 Key object 1: abnormal social-media sentiment around the announcement

For acquirer i and deal-announcement date normalized to $t = 0$, let $J_{i,[a,b]}$ denote the set of StockTwits messages about the acquirer posted between days a and b relative to the announcement. Let $\text{Sentiment}_{i,j(t)}$ denote the sentiment of tweet j about firm i posted at relative day t. The paper defines *abnormal sentiment* as the difference between average sentiment in the immediate post-announcement window and a benchmark pre-announcement window:

$$\text{AbnSent}_i \equiv \frac{1}{|J_{i,[0,3]}|} \sum_{j \in J_{i,[0,3]}} \text{Sentiment}_{i,j(t)} - \frac{1}{|J_{i,[-13,-7]}|} \sum_{j \in J_{i,[-13,-7]}} \text{Sentiment}_{i,j(t)}. \tag{1}$$

The benchmark window is intentionally not the few days immediately preceding the announcement (to mitigate leakage/anticipation). Empirically, the paper uses a standardized version (a z-score across deals), denoted Abn. Sentiment (z).

3.4 Key object 2: market-based and traditional-information controls

The paper controls for multiple alternative signals that could explain withdrawals:

- **Announcement abnormal returns:** cumulative abnormal returns (CAR) in windows around the announcement, e.g. $[-1, 10]$ and $[-5, -1]$, using Fama–French three-factor abnormal returns.
- **News media:** RavenPack sentiment and number of news articles about the deal.
- **Analysts:** net changes in recommendations and the number of recommendations.

3.5 A helpful diagnostic: AbnSent is not redundant with CAR/news/analyst signals (Table I, Panel B)

Table I, Panel B shows that "positive" vs. "negative" AbnSent does not perfectly line up with other signals. For example:

- Conditional on **negative** acquirer CAR, AbnSent is negative in 49% and positive in 51% of deals.
- Conditional on **positive** acquirer CAR, AbnSent is negative in 45% and positive in 55% of deals.

This supports the idea that AbnSent contains additional information rather than merely echoing price or news.

4 Models and estimating equations (the "core math")

4.1 Model 1: baseline deal-withdrawal regression (Linear Probability Model)

Let $\text{DealWithdrawn}_i \in \{0,1\}$ indicate whether an announced deal is ultimately withdrawn. The baseline specification is a linear probability model (LPM) that relates deal withdrawal to abnormal social-media sentiment and to a rich set of controls:

$$\begin{aligned}\text{DealWithdrawn}_i = {} & \beta_1\,\text{AbnSent}_i + \beta_2\,\text{CAR}_{i,[-1,10]} + \beta_3\,\text{CAR}_{i,[-5,-1]} + \beta_4\,\text{NewsSent}_i \\ & + \beta_5\,\Delta\text{AnalystRec}_i + \beta_6\,\text{NAnalystRec}_i + \beta_7\,\text{NTweets}_i + \beta_8\,\text{NNews}_i \\ & + \Gamma' X_i + \alpha_t + \gamma_j + \varepsilon_i. \end{aligned} \tag{2}$$

Here, X_i includes deal and firm controls (e.g., deal value, hostility, competing bidders, termination fees, size/leverage/cash of the acquirer). α_t are year-by-quarter fixed effects and γ_j are industry fixed effects (GIC 2-digit for the acquirer). In the paper, the dependent variable is multiplied by 100 for readability, so coefficients are in *percentage points*.

Interpretation of β_1. A negative β_1 means that *more positive* abnormal social-media sentiment predicts *lower* withdrawal probability (higher completion likelihood). Equivalently, negative sentiment predicts withdrawal.

4.2 Model 2: outcome valuation regression (BHAR) and interaction with withdrawal

To assess whether social media "helps" in the sense of identifying bad deals, the paper studies how the market responds over the full interim period, using buy-and-hold abnormal returns (BHAR) from $t = +11$ to deal conclusion. The regression is:

$$\text{BHAR}_{i,[11;T_{\text{concl}}]} = \beta_1 \nVdash(\text{Withdrawn})_i + \beta_2\,\text{AbnSent}_i + \beta_3 \nVdash(\text{Withdrawn})_i \times \text{AbnSent}_i + \Psi' X_i + \epsilon_i. \tag{3}$$

Interpretation of β_3. If $\beta_3 < 0$, then lower AbnSent (more negative sentiment) implies that withdrawal (vs. completion) is associated with *higher* interim BHAR—that is, the market "celebrates" withdrawing deals that social media disliked.

4.3 Model 3: corporate social-media engagement as a moderator (learning mechanism)

To test whether managers become more responsive to social-media feedback when they have better access to it, the paper uses a specification that interacts AbnSent with a post-adoption indicator for the acquirer's corporate Twitter account:

$$\text{DealWithdrawn}_i = \beta_1\,\text{AbnSent}_i + \beta_2\,\text{AbnSent}_i \times \text{Post}_{i,t} + \beta_3\,\text{Post}_{i,t} + \Omega' X_i + \alpha_t + \gamma_j + \varepsilon_i, \tag{4}$$

where $\text{Post}_{i,t} = 1$ if the deal is announced after the firm registers/activates its corporate Twitter account. With firm fixed effects, this becomes a within-firm comparison (pre vs. post adoption).

4.4 Additional "models" used to construct variables (text and classification)

Besides regressions, the paper relies on several statistical/text models to generate alternative sentiment measures and content classifications. Keeping these in the notes because they matter for understanding Tables III and VI:

Maximum-entropy classifier (logistic / softmax). A maximum-entropy sentiment classifier typically estimates

$$\Pr(y = 1 \mid x) = \frac{\exp(\theta' x)}{1 + \exp(\theta' x)},$$

where x is a vector of text features (e.g., word indicators) and y is a bullish/bearish label. This is used to impute sentiment for tweets without explicit labels.

Naive Bayes classifier. A standard naive Bayes text classifier uses

$$\Pr(y \mid x) \propto \Pr(y) \prod_k \Pr(x_k \mid y),$$

with conditional-independence assumptions across features x_k.

Biterm topic model (BTM). BTM is designed for short texts (tweets). It models co-occurring word pairs (biterms) rather than full document-level mixtures as in LDA. The paper uses BTM to identify tweet topics (e.g., deal-terms, company/business, memes), then constructs topic-specific AbnSent signals (Table VI, Panel B).

5 Reading the tables and figures (what each result is telling me)

5.1 Table I: summary statistics and basic patterns

Panel A (completed vs. withdrawn). Withdrawn deals differ systematically: they have (i) *more negative* AbnSent and CAR, (ii) much *longer deal delay*, (iii) *larger deal value*, and (iv) more indicators of conflict/complexity (competing bidders, challenged/rumored/hostile deals). Interestingly, withdrawn deals have *more news articles* on average, consistent with salience and scrutiny.

5

Panel B (signal non-redundancy). AbnSent disagrees with CAR/news/analyst signals in a large fraction of deals (roughly close to 50/50 in many splits). This motivates the multivariate approach in Table II: AbnSent is not merely a proxy for price or news reactions.

5.2 Figures 2–4: social media attention reacts sharply to announcements

Figures 2 and 3 show that StockTwits activity spikes strongly at $t = 0$ and $t = 1$ around merger announcements, and that posts about the acquirer that also mention the target ticker spike as well (for public targets). Figure 4 shows that the *content* of tweets shifts toward M&A language at the announcement. I interpret these figures as validating the use of short post-announcement windows $[0, 3]$ as capturing an announcement reaction that was not fully anticipated.

Table I

Summary Statistics

This table presents summary statistics for the main variables in our sample. Panel A compares the variables across completed mergers and withdrawn mergers. The last three columns of the table present the average difference, standard error, and corresponding *p*-value of the average difference. All variables are defined as detailed in Section I and the Appendix. All M&A deal characteristics are obtained from SDC Platinum, all accounting variables are obtained from Compustat North America and winsorized at the 5% level within the full Compustat universe. Stock returns and cumulative abnormal returns are constructed using data from CRSP and the Fama-French three-factor model as detailed in Section I.B. News media sentiment and coverage data are from RavenPack. The sample covers all M&A deals with U.S. acquiring firms over the period from 2010 to 2021 with a minimum deal volume of $25M. Panel B cross-tabulates positive (i.e., greater than zero) and negative abnormal sentiment around M&A announcements with positive and negative stock market reactions (CAR), news sentiment reactions, and analyst recommendation changes. Summary statistics for the full set of variables in the paper are presented in Internet Appendix Table IA.II.

Panel A: Completed versus Withdrawn Mergers

	Completed ($N = 6,187$)		Withdrawn ($N = 251$)				
Variable	Mean	*SD*	Mean	*SD*	Diff.	SE	*p*-Value
Abn. Sent (StTw)	0.022	0.305	−0.030	0.271	−0.052	0.018	0.003
Abn. Sent (MaxE)	0.038	0.166	−0.009	0.151	−0.047	0.010	0.000
Abn. Sent (SMA-Tw)	0.297	0.590	0.350	0.623	0.053	0.047	0.255
CAR Acq. [−1,10]	0.008	0.090	−0.016	0.093	−0.024	0.006	0.000
News Sent. Acq.	0.007	0.997	−0.181	1.046	−0.189	0.067	0.005
N Tweets	193.316	1961.903	148.036	519.080	−45.280	41.178	0.272
N News Articles	4.500	17.014	8.888	18.126	4.389	1.164	0.000
Deal Delay	80.350	106.566	179.649	187.172	99.299	11.892	0.000
Deal Value (B. USD)	1.028	2.517	4.466	6.030	3.438	0.382	0.000
Pct. Held Prior	2.157	10.867	2.921	11.471	0.765	0.737	0.301
Acq. White Knight (0/1)	0.001	0.025	0.012	0.109	0.011	0.007	0.102
Competing Bidder (0/1)	0.008	0.088	0.211	0.409	0.203	0.026	0.000
Challenged Deal (0/1)	0.009	0.096	0.275	0.447	0.266	0.028	0.000
Rumored Deal (0/1)	0.093	0.291	0.207	0.406	0.114	0.026	0.000
Target Private (0/1)	0.419	0.494	0.120	0.325	−0.300	0.021	0.000
Hostile Deal (0/1)	0.000	0.013	0.064	0.245	0.064	0.015	0.000
Pct. Shares Sought	96.714	13.018	95.906	14.497	−0.807	0.941	0.392
Target Termination Fee	15.564	101.350	37.422	172.388	21.858	10.957	0.047

Panel B: Cross-Tabulation of Positive and Negative M&A Announcement Reactions

	AbnSent	
	Negative	Positive
CAR Acq.		
Negative	1,383 (49%)	1,423 (51%)
Positive	1,413 (45%)	1,713 (55%)
News Sent.		
Negative	459 (49%)	472 (51%)
Positive	2,337 (47%)	2,664 (53%)
Analyst Recd.		
Negative	2,293 (48%)	2,507 (52%)
Positive	503 (44%)	629 (56%)

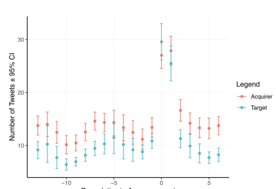

Figure 2. Number of tweets around merger announcements. This figure shows the average and 95% confidence interval of the number of tweets posted to StockTwits around the announcement of an M&A deal that mention the acquiring and target firm, respectively. The solid line displays the numbers for the acquirer firm; the dashed line represents the target firm. The number of tweets about the target firm is available only when the target firm is public, that is, in StockTwits. (Color figure can be viewed at wileyonlinelibrary.com)

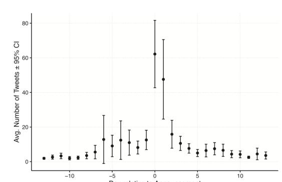

Figure 3. Tweets about acquirer that include target ticker. This figure shows the average and 95% confidence interval of the number of tweets about the acquiring firm posted to StockTwits around the announcement of an M&A deal that also include the ticker of the target firm, relative to the deal announcement date. The sample underlying this figure only includes deals with publicly listed target firms, as private firms do not have a "cashtag" in StockTwits.

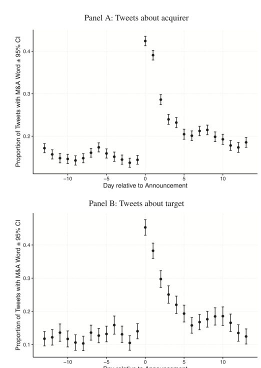

Figure 4. M&A words in tweets. This figure shows the proportion of tweets that include M&A-related words (relative to the total number of tweets) around the announcement of an M&A deal about the acquirer (Panel A) and target firm (Panel B). Panel A displays the average and 95% confidence interval of the proportion of tweets posted to StockTwits that include at least one word related to mergers and acquisitions (i.e., "merger," "acquisition," "m&a," "takeover," "acquirer," and "target"). Panel B plots the corresponding numbers for the target firm.

5.3 Table II: baseline effect of AbnSent on withdrawal probability

Panel A. Across specifications, AbnSent is a robust negative predictor of withdrawal. In the simplest regression (column (1)), a one-standard-deviation increase in AbnSent is associated with a *0.643 percentage point* lower probability of withdrawal (because the dependent variable is scaled by 100). Given a mean withdrawal rate of about 3.885%, this is economically meaningful:

$$\frac{0.643}{3.885} \approx 16.6\% \quad \text{of the unconditional withdrawal probability.}$$

The coefficient remains similar after adding CAR controls, news sentiment and volume, analyst recommendation changes, and rich deal characteristics (columns (2)–(6)).

Interpretation. The reading is that AbnSent contains incremental information about whether a deal will be withdrawn that is not absorbed by:

- short-window market reactions (CAR),
- traditional news sentiment/coverage,
- or analyst recommendation changes.

This is the paper's main "social media is informative for corporate decisions" result.

Panel B (StockTwits vs. Twitter sentiment). When the authors use Twitter sentiment from a vendor (SMA) and include both signals, both are negative and significant, which suggests (i) the result is not platform-specific and (ii) there is non-trivial independent variation across platforms.

Table II

Social Media Feedback and M&A Outcomes

This table presents linear probability model estimates for the effect of social media feedback on the likelihood of M&A deal withdrawal. In Panels A and B, the dependent variable is an indicator variable taking the value of one if the announced M&A transaction was subsequently withdrawn, and zero otherwise. For legibility, we multiply the dependent variable by 100 in all regressions. *Abn. Sentiment (z) (StTw)* is the social media feedback from StockTwits, constructed as the difference between average StockTwits sentiment around the M&A deal announcement ([0; 3]) and a benchmark period before the announcement. *Abn. Sentiment (z) (SMA-Tw)* is constructed similarly using Twitter sentiment data from SMA. All variables denoted with "(z)" are standardized to have a mean of zero and a standard deviation of one. *CAR Acq. (z) [−1;10] ([−5;−1])* are the cumulative abnormal returns of the acquiring firm in the [−1; 10] and [−5; −1] window around the M&A announcement. *News Sentiment Acq. (z)* and *N News Articles* are the news media sentiment and the number of news articles published about the M&A deal, respectively, from RavenPack. *Analyst Rec. Changes* and *N Analyst Rec.* are the net changes and the number of analyst recommendations, and *N Tweets* is the number of StockTwits tweets around the deal announcement. All of the other M&A deal characteristics are from SDC Platinum; detailed variable definitions are provided in the Appendix. *Mean(LHS)* is the average of the dependent variable in the given regression. Acquirer controls (firm size, leverage, and cash holdings) are included as indicated. All regressions include year-by-quarter and acquiring-firm industry (GIC two-digit) fixed effects. Standard errors are clustered at the year-by-quarter level and are reported in parentheses. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% level, respectively.

Panel A: Baseline Effect

	𝟙*(Deal Withdrawn)*					
	(1)	(2)	(3)	(4)	(5)	(6)
Abn. Sentiment (z) (StTw)	$\boxed{-0.6434^{***}}$	−0.7712***	−0.7153***	−0.7165***	−0.7294***	−0.7110***
	(0.2185)	(0.2167)	(0.2165)	(0.2066)	(0.2067)	(0.2099)
CAR Acq. (z) [−1;10]			−0.8760***	−0.9090***	−0.9259***	−0.8353***
			(0.2620)	(0.2562)	(0.2563)	(0.2689)
CAR Acq. (z) [−5;−1]			0.6584**	0.4687*	0.4713*	0.4688*
			(0.2601)	(0.2614)	(0.2663)	(0.2630)
News Sentiment Acq. (z)					−1.024***	−1.019***
					(0.2633)	(0.2578)
Analyst Rec. Changes (z)						−0.9653***
						(0.3246)
N Analyst Rec.						0.0986
						(0.1898)
N News Articles					0.0092	0.0094
					(0.0102)	(0.0101)
Log Deal Value ($B)		8.275***	8.262***	7.795***	8.009***	8.115***
		(0.7574)	(0.7667)	(0.8927)	(0.9041)	(0.9721)
% Shares Held Prior		0.0350	0.0324	0.0434*	0.0380	0.0385
		(0.0228)	(0.0236)	(0.0228)	(0.0232)	(0.0232)
Acq. White Knight (0/1)				0.5101	0.2363	−0.2203
				(16.27)	(16.21)	(16.29)
Competing Bidder (0/1)				43.02***	42.89***	42.77***
				(5.341)	(5.340)	(5.296)
Rumored Deal (0/1)				−1.196	−1.174	−1.196
				(1.144)	(1.151)	(1.136)
Hostile Deal (0/1)				76.96***	76.16***	76.16***
				(6.683)	(6.530)	(6.448)

(Continued)

Table II—*Continued*

Panel A: Baseline Effect

	𝟙*(Deal Withdrawn)*					
	(1)	(2)	(3)	(4)	(5)	(6)
Termination Fee Target ($M)				−0.0167***	−0.0171***	−0.0165***
				(0.0045)	(0.0045)	(0.0044)
N Tweets					−0.0002***	−0.0002***
					(0.00004)	(0.00004)
Mean(LHS)	3.885	3.763	3.776	3.776	3.776	3.776
Observations	6,306	5,979	5,932	5,932	5,932	5,932
R^2	0.0011	0.0710	0.0738	0.2107	0.2137	0.2161
Firm Controls		✓	✓	✓	✓	✓
Year-by-Quarter FE		✓	✓	✓	✓	✓
Acq. Industry (GIC2) FE		✓	✓	✓	✓	✓

Panel B: StockTwits versus Twitter Sentiment

	𝟙*(Deal Withdrawn)*		
	(1)	(2)	(3)
Abn. Sentiment (z) (StTw)	−0.7060**		−0.5925**
	(0.2796)		(0.2839)
Abn. Sentiment (z) (SMA-Tw)		−1.075***	−0.9981***
		(0.2705)	(0.2804)
CAR Acq. (z) [−1;10]	−1.145***	−0.9775***	−0.9609***
	(0.2713)	(0.2748)	(0.2706)
CAR Acq. (z) [−5;−1]	0.4592	0.4610	0.4504
	(0.3158)	(0.3125)	(0.3117)
News Sentiment Acq. (z)	−1.330***	−1.253***	−1.266***
	(0.2969)	(0.2955)	(0.2948)
Log Deal Value ($B)	7.672***	8.176***	8.151***
	(0.9369)	(0.9439)	(0.9433)
% Shares Held Prior	0.0346	0.0322	0.0326
	(0.0335)	(0.0331)	(0.0329)
Acq. White Knight (0/1)	−8.539	−9.037	−8.838
	(16.50)	(16.28)	(16.37)
Competing Bidder (0/1)	42.64***	42.62***	42.62***
	(6.074)	(6.029)	(6.036)
Rumored Deal (0/1)	−0.6059	−0.5472	−0.5514
	(1.410)	(1.431)	(1.423)
Hostile Deal (0/1)	79.80***	78.95***	79.07***
	(5.116)	(5.265)	(5.284)
Termination Fee Target ($M)	−0.0150***	−0.0145***	−0.0145***
	(0.0045)	(0.0044)	(0.0045)
N Tweets	−0.0002	−0.0001	−0.0002
	(0.0003)	(0.0003)	(0.0003)
N News Articles	0.0023	0.0024	0.0023
	(0.0082)	(0.0080)	(0.0079)
Mean(LHS)	3.864	3.885	3.756
Observations	4,553	4,553	4,553
R^2	0.2203	0.2215	0.2223
Firm Controls	✓	✓	✓
Year-by-Quarter FE	✓	✓	✓
Acq. Industry (GIC2) FE	✓	✓	✓

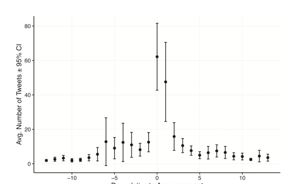

Figure 3. Tweets about acquirer that include target ticker. This figure shows the average and 95% confidence interval of the number of tweets about the acquiring firm posted to StockTwits around the announcement of an M&A deal that also include the ticker of the target firm, relative to the deal announcement date. The sample underlying this figure only includes deals with publicly listed target firms, as private firms do not have a "cashtag" in StockTwits.

5.4 Figure 5: the effect is not limited to negative-CAR deals

Figure 5 shows that AbnSent predicts withdrawal in both negative and nonnegative announcement-CAR subsamples. This helps me rule out a simple story that AbnSent is only informative because it mechanically mirrors negative stock-price reactions.

5.5 Table III: robustness and separating learning from "pure forecasting"

Panel A: alternative sentiment measures and alternative samples/specs.

- Using alternative imputed sentiment measures (maximum entropy or naive Bayes) still yields a negative relation to withdrawal.
- In public-target deals, AbnSent remains predictive even after controlling for target-side reactions (target sentiment and target CAR). The target-side AbnSent does not add predictive power beyond the acquirer's AbnSent.
- Adding the *target spread* (offer price vs. target stock price) increases explanatory power but does not eliminate the AbnSent effect. I interpret this as evidence that AbnSent is not just picking up "probability of completion" as inferred from arbitrage spreads.
- With **acquirer fixed effects**, the AbnSent coefficient remains negative and significant. This is important because it uses within-firm variation: the same acquirer can have deals that social media likes more or less, and that variation predicts the likelihood of withdrawal.

Panel B: external withdrawal reasons. When the sample is restricted to deals withdrawn due to regulators or due to the target (shareholders/board), the AbnSent coefficient is much smaller in magnitude (and sometimes insignificant). I read this as supporting the "manager learning" channel: social media should predict withdrawals most strongly when the acquirer can choose to walk away, not when the outcome is forced by external parties.

Table III
Robustness: Social Media Feedback and M&A Outcomes

This table presents linear probability model estimates for the effect of social media feedback on the likelihood of M&A deal withdrawal, analogous to Table II. The dependent variable is a dummy variable indicating whether an announced M&A transaction was subsequently withdrawn, multiplied by 100. In Panel A, *Abn. Sentiment (z) (MaxE)* and *Abn. Sentiment (z) (Bayes)* are the social media feedback from StockTwits using the maximum entropy and naive Bayes classifier, respectively, constructed similarly as *Abn. Sentiment (z) (StTw)* in Table II. *Abn. Sent. Target (z) (StTw)* is the corresponding social media feedback for the target firm, constructed using StockTwits sentiment data. *CAR Target (z) [−1;10] ([−5;−1])* are the cumulative abnormal returns of the target firm in the [−1; 10] and [−5; −1] windows around the M&A announcement. In Panel B, we only include M&A deals that were either completed or that were withdrawn for the indicated reason (i.e., due to regulators in column (1) and due to target shareholders or the target firm board in column (2)). All other variables are similar to those in Table II. Each regression includes similar deal and firm-level controls as Table II and year-by-quarter and acquiring-firm industry (GIC two-digit) fixed effects as indicated. Standard errors are clustered at the year-by-quarter level and are reported in parentheses. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% level, respectively.

Panel A: Alternative Measures and Specifications

	$\mathbb{1}$*(Deal Withdrawn)*							
Specification	MaxEnt	Bayes	Public Targets				Drop Small	Acq. FE
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Abn. Sentiment (z) (MaxE)	−0.5835***							
	(0.1971)							
Abn. Sentiment (z) (Bayes)		−0.4268**						
		(0.2053)						
Abn. Sentiment (z) (StTw)			−2.133**	−2.962**	−2.804*	−2.607*	−1.170***	−0.8011**
			(0.9750)	(1.438)	(1.447)	(1.492)	(0.2785)	(0.3337)
Abn. Sent. Target (z) (StTw)				0.1704	0.0286	0.0228		
				(1.119)	(1.160)	(1.167)		
CAR Acq. (z) [−1;10]	−0.9461***	−0.9609***	−1.443	−2.042*	−1.949	−1.129	−1.315***	−1.278***
	(0.2501)	(0.2560)	(1.178)	(1.213)	(1.189)	(1.240)	(0.2813)	(0.3612)
CAR Acq. (z) [−5;−1]	0.5323**	0.5319**	1.064	0.7762	1.176	0.9680	0.6810**	0.7815**
	(0.2598)	(0.2593)	(1.154)	(1.280)	(1.328)	(1.421)	(0.3218)	(0.3277)
CAR Target (z) [−1;10]			−1.230		−1.571	−1.921		
			(1.148)		(1.245)	(1.440)		
CAR Target (z) [−5;−1]			−0.7008		−0.5146	−0.7775		
			(1.146)		(1.183)	(1.121)		

(Continued)

Table III—*Continued*

Panel A: Alternative Measures and Specifications

	$\mathbb{1}$*(Deal Withdrawn)*							
Specification	MaxEnt	Bayes	Public Targets				Drop Small	Acq. FE
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
News Sentiment Acq. (z)	−1.007***	−1.004***	−3.932**	−4.022***	−4.036***	−2.440*	−1.612***	−1.049***
	(0.2566)	(0.2579)	(1.500)	(1.433)	(1.461)	(1.381)	(0.3662)	(0.3757)
N Tweets	−0.0002***	−0.0002***	−0.0006	−0.0008*	−0.0007	−0.0013***	−0.0002***	0.0002
	(0.00004)	(0.00004)	(0.0005)	(0.0004)	(0.0004)	(0.0005)	(0.00004)	(0.0006)
N News Articles	0.0087	0.0089	0.0551	0.0272	0.0353	0.0112	0.0102	0.0129
	(0.0102)	(0.0103)	(0.0636)	(0.0585)	(0.0566)	(0.0461)	(0.0103)	(0.0120)
Target Spread						20.23*		
						(11.54)		
Mean(LHS)	3.791	3.791	4.139	11.57	11.63	10.91	3.914	3.825
Observations	5,988	5,988	880	683	679	605	4,445	5,896
R^2	0.2117	0.2113	0.3372	0.3523	0.3567	0.3639	0.2344	0.5536
Deal Controls	✓	✓	✓	✓	✓	✓	✓	✓
Firm Controls	✓	✓	✓	✓	✓	✓	✓	✓
Year-by-Quarter FE	✓	✓	✓	✓	✓	✓	✓	✓
Acq. Industry (GIC2) FE	✓	✓	✓	✓	✓	✓	✓	✓
Acq. Firm FE								✓

Table III—*Continued*

Panel B: Deal Rejection Reasons

	$\mathbb{1}$*(Deal Withdrawn)*	
Deal Rejected by:	Regulator (1)	Target (2)
Abn. Sentiment (z) (StTw)	−0.2020**	−0.1504
	(0.0883)	(0.1219)
CAR Acq. (z) [−1;10]	−0.0226	−0.2921***
	(0.0725)	(0.0974)
CAR Acq. (z) [−5;−1]	0.2586**	0.0806
	(0.1139)	(0.1168)
News Sentiment Acq. (z)	−0.2009*	−0.8442***
	(0.1141)	(0.2059)
N Tweets	-3.81×10^{-5}**	-7.73×10^{-5}***
	(1.48×10^{-5})	(1.88×10^{-5})
N News Articles	0.0041	0.0038
	(0.0080)	(0.0060)
N Deals Withdr.	30	74
Observations	5,734	5,776
R^2	0.0564	0.2790
Deal Controls	✓	✓
Firm Controls	✓	✓
Year-by-Quarter FE	✓	✓
Acq. Industry (GIC2) FE	✓	✓

5.6 Table IV: does the market *like* the withdrawals that social media predicts?

Table IV estimates equation (3). The key term is the interaction AbnSent $\times\, \mathbb{1}$(Withdrawn). It is significantly negative, implying:

> When social media sentiment at announcement is *more negative*, the eventual withdrawal (rather than completion) is associated with *better* interim buy-and-hold abnormal returns.

This strengthens the interpretation that AbnSent is informative about *deal quality*: negative sentiment identifies deals whose abandonment is value-improving.

5.7 Table V: corporate Twitter accounts and increased sensitivity to AbnSent

Table V implements the "engagement" test. The interaction between AbnSent and a generic corporate Twitter indicator is negative but not always significant. However, interactions with *high-follower* accounts and *verified* accounts are economically larger and statistically significant. I interpret this as: mere presence on Twitter is not enough; the mechanism appears stronger when the corporate account is *active/credible* and therefore plausibly used to monitor (or to be exposed to) social discourse.

Table IV
Social Media Feedback and Deal BHARs

This table summarizes OLS regressions of the buy-and-hold abnormal returns (BHAR) from M&A deal announcement to deal conclusion on the initial social media and stock market feedback, the eventual deal outcome (i.e., completion or withdrawal), and their interactions. The dependent variable, *Acq. BHAR*$[11; T_{conclusion}]$, is constructed over the event window from 11 days after the deal announcement until deal conclusion ($T_{conclusion}$), using abnormal returns with respect to the Fama-French three-factor model. As before, *Abn. Sentiment (z) (StTw)* is the social media feedback from StockTwits, and *CAR Acq. (z) [−1;10] ([−5;−1])* are the cumulative abnormal returns of the acquiring firm in the [−1; 10] and [−5; −1] window around the M&A announcement. $\mathbb{1}$(*Deal Withdr.*) is an indicator variable that takes the value of one if the deal was ultimately withdrawn and zero if it was completed. To avoid an overlap between the event window around the deal announcement and the deal conclusion, columns (1) and (2) retain only deals with a minimum delay between announcement and conclusion of at least 25 days, and columns (3) and (4) retain only deals with a delay of more than 75 days. In all regressions, we exclude deals that were rejected by the target firm. All other variables are similar to those in Table II. Each regression includes similar deal and firm-level controls as Table II and year-by-quarter and acquiring-firm industry (GIC two-digit) fixed effects as indicated. Standard errors are clustered at the year-by-quarter level and are reported in parentheses. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% level, respectively.

	BHAR Acq. [+11; Deal Conclusion]			
	Delay >25 days		Delay >75 days	
	(1)	(2)	(3)	(4)
Abn. Sentiment (z) (StTw) × $\mathbb{1}$*(Deal Withdr.)*	−0.1009**	−0.1087**	−0.1427***	−0.1528***
	(0.0390)	(0.0423)	(0.0309)	(0.0357)
Abn. Sentiment (z) (StTw)	−0.0078	−0.0074	−0.0154*	−0.0150
	(0.0050)	(0.0050)	(0.0090)	(0.0090)
$\mathbb{1}$*(Deal Withdr.)* × *CAR Acq. (z) [−1;10]*		0.1505		0.1710
		(0.1018)		(0.1347)
$\mathbb{1}$*(Deal Withdr.)* × *CAR Acq. (z) [−5;−1]*		0.0212		0.0319
		(0.0449)		(0.0558)
$\mathbb{1}$*(Deal Withdr.)*	−0.0608	−0.0242	−0.0715	−0.0328
	(0.0622)	(0.0785)	(0.0820)	(0.1018)
CAR Acq. (z) [−1;10]	0.0389***	0.0324***	0.0624***	0.0519***
	(0.0114)	(0.0100)	(0.0204)	(0.0181)
CAR Acq. (z) [−5;−1]	0.0116	0.0096	0.0211	0.0175
	(0.0118)	(0.0116)	(0.0218)	(0.0218)
News Sentiment Acq. (z)	0.0115**	0.0118**	0.0233**	0.0226**
	(0.0051)	(0.0051)	(0.0096)	(0.0093)
Mean(LHS)	3.857	3.857	3.890	3.890
Observations	3,343	3,343	1,784	1,784
R^2	0.0444	0.0523	0.0781	0.0864
(Soc) Media Controls	✓	✓	✓	✓
Deal Controls	✓	✓	✓	✓
Firm Controls	✓	✓	✓	✓
Year-by-Quarter FE	✓	✓	✓	✓
Acq. Industry (GIC2) FE	✓	✓	✓	✓

Table V
Acquirers' Social Media Engagement

This table presents linear probability model estimates on the effect of corporate social media engagement on the relationship between M&A deal completion and social media feedback around M&A deal announcement. Similar to Table II, the dependent variable is an indicator that takes the value of one if the previously announced M&A deal is ultimately withdrawn, and zero otherwise, and *Abn. Sentiment (z) (StTw)* is the social media feedback from StockTwits. $\mathbb{1}$(*Acq. Twitter*) is an indicator variable that takes the value of one for firms that have a corporate Twitter account that was registered and activated before the announcement of the M&A deal, and zero otherwise. Similarly, $\mathbb{1}$(*Acq. HighFollow*) and $\mathbb{1}$(*Acq. Verified*) are indicator variables that take the value of one if at the time of deal announcement the acquiring firm had a corporate Twitter account with an above-median number of followers and with "verified account" (i.e., blue check-mark) status, respectively. All other variables are similar to those in Table II. Each regression includes similar deal and firm-level controls as Table II and year-by-quarter and acquiring-firm industry (GIC two-digit) fixed effects as indicated. Standard errors are clustered at the year-by-quarter level and are reported in parentheses. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% level, respectively.

	$\mathbb{1}$*(Deal Withdrawn)*					
	(1)	(2)	(3)	(4)	(5)	(6)
Abn. Sentiment (z) (StTw) × $\mathbb{1}$*(Acq. Twitter)*	−0.4711	−0.4472				
	(0.4025)	(0.4206)				
Abn. Sentiment (z) (StTw) × $\mathbb{1}$*(Acq. HighFollow)*			−0.9646**	−0.9559**		
			(0.4352)	(0.4608)		
Abn. Sentiment (z) (StTw) × $\mathbb{1}$*(Acq. Verified)*					−1.301**	−0.9259*
					(0.5019)	(0.5262)
Abn. Sentiment (z) (StTw)	−0.4356*	−0.4998	−0.3400*	−0.3764	−0.4000**	−0.5430
	(0.2495)	(0.3752)	(0.2002)	(0.3025)	(0.1981)	(0.3277)
$\mathbb{1}$*(Acq. Twitter)*	0.4215	−0.8588				
	(0.5940)	(1.766)				
$\mathbb{1}$*(Acq. HighFollow)*			0.4742	−1.722		
			(0.6047)	(3.602)		
$\mathbb{1}$*(Acq. Verified)*					0.7672	−3.835
					(0.6698)	(3.799)

Table V—*Continued*

	$\mathbb{1}$*(Deal Withdrawn)*					
	(1)	(2)	(3)	(4)	(5)	(6)
Mean(LHS)	3.776	3.776	3.776	3.776	3.776	3.776
Observations	5,932	5,932	5,932	5,932	5,932	5,932
R^2	0.2139	0.5538	0.2144	0.5541	0.2148	0.5542
Deal Controls	✓	✓	✓	✓	✓	✓
Firm Controls	✓	✓	✓	✓	✓	✓
(Social) Media Controls	✓	✓	✓	✓	✓	✓
CAR Controls	✓	✓	✓	✓	✓	✓
Year-by-Quarter FE	✓	✓	✓	✓	✓	✓
Acq. Industry (GIC2) FE	✓	✓	✓	✓	✓	✓
Acq. Firm FE		✓		✓		✓

5.8 Table VI and Figure 6: who generates the information in social media?

Table VI shows that the signal is *not* driven by noisy content:

- **Nontechnical tweets** predict withdrawal strongly; **technical-trader tweets** do not.
- **Long tweets** (above-median word count) predict withdrawal strongly; short tweets do not.
- Topic-specific sentiment is informative for **fundamental topics** (company/business, deal terms, disclosure), but not for memes or technical content.

Figure 6 complements this: around announcements, fundamental users both tweet more and use more M&A-related language. Together, I read these results as a "content validity" check: the predictive AbnSent signal seems to come from more analysis-rich investor commentary.

5.9 Table VII: relationship to other information sources

Table VII splits the sample by social-media intensity (number of tweets), traditional-media intensity (number of news articles), and the magnitude of announcement CAR:

- The AbnSent coefficient is much stronger when there are **many StockTwits messages** around the announcement (signal quality improves with more social media information).
- The AbnSent coefficient does *not* differ much across high- vs. low-news-coverage subsamples (traditional news volume does not explain the effect).
- The AbnSent coefficient is stronger when |CAR| is high, consistent with the signal being stronger for more salient/controversial deals.

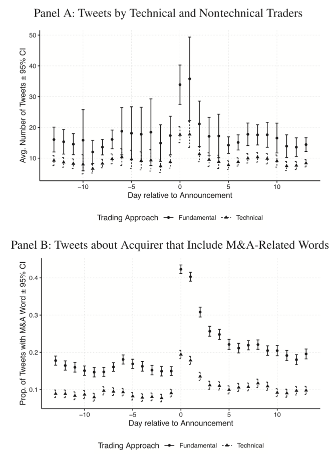

Figure 6. Tweets by (non)technical traders. This figure shows the dynamics of StockTwits posts around M&A announcements for technical and fundamental StockTwits users. We classify individual tweets posted to StockTwits into "technical" and "fundamental" (i.e., nontechnical) tweets using a maximum entropy classifier trained on the messages of users who declare that they use a technical approach to trading. Panel A shows the average and 95% confidence interval of the number of tweets around the M&A announcement. Panel B shows the average proportion and 95% confidence interval of tweets that contain words related to mergers and acquisitions (relative to the total number of tweets), similar to Figure 4. Each figure plots the dynamics separately for "technical" and "other" (i.e., nontechnical) posts.

Table VI

Nature of StockTwits Content

This table presents linear probability model estimates on the effect of social media feedback on M&A deal completion focusing on the nature of the content of the tweets posted to StockTwits. Panel A focuses on technical posts and tweet length: to construct *Abn. Sent (z) Technical=N* and *Abn. Sent (z) Technical=Y* in columns (1) and (2), we classify each StockTwits post as technical or nontechnical content using a maximum entropy classifier trained on the tweets of users who declare themselves as technical traders (i.e., chart formations and patterns, etc.). We then aggregate the tweets across these two subgroups and construct *Abn. Sentiment* as before. Similarly, *Abn. Sent (z) Long=Y* (*Abn. Sent (z) Long=N*) are constructed by separately estimating abnormal sentiment around merger announcement for tweets with above- and below-median number of words (i.e., seven words) in the tweet. Panel B focuses on abnormal sentiment from subsamples of tweets based on the topic of the tweet. We use a biterm topic modeling (BTM) approach to identify the topic of a given tweet and construct *AbnSent* for each subset of tweets individually, similar to Panel A. All variables denoted with "(z)" are standardized to have a mean of zero and a standard deviation of one. All other variables are similar to those in Table II. Each regression includes similar deal and firm-level controls as Table II and year-by-quarter and acquiring-firm industry (GIC two-digit) fixed effects as indicated. Standard errors are clustered at the year-by-quarter level and are reported in parentheses. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% level, respectively.

Panel A: Abnormal Sentiment by Tweet Type and Length

	𝟙*(Deal Withdrawn)*			
	Technical Traders		Tweet Length	
	(1)	(2)	(3)	(4)
Abn. Sent (z) Technical = N	−1.105***	−1.080***		
	(0.2905)	(0.2884)		
Abn. Sent (z) Technical = Y	−0.3497	−0.2744		
	(0.3095)	(0.3010)		
Abn. Sent (z) Long = Y			−1.214***	−1.222***
			(0.3033)	(0.2983)
Abn. Sent (z) Long = N			−0.0525	0.0140
			(0.2664)	(0.2616)
CAR Acq. (z) [−1;10]		−1.023***		−0.8895***
		(0.3024)		(0.2837)
CAR Acq. (z) [−5;−1]		0.3161		0.4885
		(0.2983)		(0.2914)
News Sentiment Acq. (z)		−1.191***		−1.202***
		(0.3365)		(0.3239)
Mean(LHS)	4.239	4.247	3.909	3.920
Observations	3,680	3,650	4,042	4,005
R^2	0.2331	0.2430	0.2272	0.2361
(Soc) Media Controls		✓		✓
Deal Controls	✓	✓	✓	✓
Firm Controls	✓	✓	✓	✓
Year-by-Quarter FE	✓	✓	✓	✓
Acq. Industry (GIC2) FE	✓	✓	✓	✓

Table VI—*Continued*

Panel B: Abnormal Sentiment by Tweet Topic

	𝟙*(Deal Withdrawn)*				
	(1)	(2)	(3)	(4)	(5)
Abn. Sent (z) Company/Business	−1.202***				
	(0.3160)				
Abn. Sent (z) Deal-Terms		−1.139***			
		(0.2983)			
Abn. Sent (z) Disclosure			−0.8675**		
			(0.3987)		
Abn. Sent (z) Meme				−0.2804	
				(0.6155)	
Abn. Sent (z) Technical					−0.4872
					(0.3391)
Abn. Sent (z) Trading	0.1189	0.1649	0.5371	−1.082	−0.3840
	(0.2733)	(0.2771)	(0.3953)	(0.7547)	(0.3508)
CAR Acq. (z) [−1;10]	−1.091***	−0.8503***	−0.9082***	−1.159***	−0.9557***
	(0.2858)	(0.2869)	(0.3209)	(0.3449)	(0.2714)
CAR Acq. (z) [−5;−1]	0.5760*	0.4322	0.4014	−0.0846	0.7333**
	(0.3007)	(0.3320)	(0.4072)	(0.4911)	(0.3166)
News Sentiment Acq. (z)	−0.9505**	−1.366***	−1.158***	−0.4856	−1.419***
	(0.3645)	(0.3482)	(0.4149)	(0.6599)	(0.3654)
Mean(LHS)	4.339	4.202	4.030	4.103	4.333
Observations	3,941	4,117	3,151	1,121	3,600
R^2	0.2450	0.2531	0.2480	0.2345	0.2546
(Soc) Media Controls	✓	✓	✓	✓	✓
Deal Controls	✓	✓	✓	✓	✓
Firm Controls	✓	✓	✓	✓	✓
Year-by-Quarter FE	✓	✓	✓	✓	✓
Acq. Industry (GIC2) FE	✓	✓	✓	✓	✓

5.10 Table VIII: analyst conference calls and the Q&A channel

Table VIII asks whether the AbnSent signal is different when acquirers hold M&A-related analyst conference calls and when the call language is more negative/constraining. The strongest takeaway for me is in Panel B:

> The AbnSent signal is significantly more predictive when the **Q&A portion** of the call contains a high fraction of constraining or negative words, while the scripted presentation portion does not show the same pattern.

This is consistent with an *external feedback* mechanism: the Q&A is where investors push back, reveal concerns, and surface constraints.

6 Identification logic and interpret causality

6.1 What the paper can and cannot claim

The paper is careful: it primarily claims that social media sentiment *predicts* withdrawals and provides evidence consistent with a *learning/feedback* mechanism. I interpret the key identification challenge as: AbnSent is not randomly assigned, so a purely causal claim "social media causes withdrawals" is difficult.

6.2 Why the learning interpretation is plausible

The combination of results that push toward a learning interpretation (rather than pure correlation) are:

1. AbnSent remains predictive *after controlling* for prices, news, and analyst signals (Table II).
2. AbnSent is less predictive when withdrawals are driven by external parties (Table III, Panel B).
3. The AbnSent effect strengthens when firms have stronger channels to listen (Twitter engagement; Table V).

Table VII

Other Information Sources: Stock Market and News Media

This table presents linear probability model estimates for the effect of social media feedback on the likelihood of M&A deal withdrawal, focusing on the relationship between social media and other sources of information. Similar to Table II, the dependent variable is a dummy variable indicating whether an announced M&A transaction was subsequently withdrawn, multiplied by 100. All explanatory variables are defined similarly as in Table II. We split the sample into observations with above- and below-median number of tweets posted on StockTwits (columns (1) and (2)), the number of news articles about the M&A deal (columns (3) and (4)), and the absolute value of the acquirer's M&A announcement returns (columns (5) and (6)), respectively. Each regression includes similar deal and firm-level controls as Table II and year-by-quarter and acquiring-firm industry (GIC two-digit) fixed effects as indicated. Standard errors are clustered at the year-by-quarter level and are reported in parentheses. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% level, respectively.

	𝟙*(Deal Withdrawn)*					
Sample Split	*N Tweets*		*N News Articles*		*Abs(CAR)*	
	Low (1)	High (2)	Low (3)	High (4)	Low (5)	High (6)
Abn. Sentiment (z) (StTw)	−0.5157*	−2.030***	−0.6450**	−0.8438*	−0.3169	−1.202***
	(0.2560)	(0.6128)	(0.2692)	(0.4818)	(0.2307)	(0.3090)
CAR Acq. (z) [−1;10]	−0.8590*	−1.100***	−0.5859	−1.726***	0.2068	−0.9857***
	(0.4697)	(0.2573)	(0.3725)	(0.4473)	(1.291)	(0.2653)
CAR Acq. (z) [−5;−1]	0.3030	0.6596*	0.5017	0.8547	0.1316	0.6140
	(0.4102)	(0.3391)	(0.3540)	(0.5448)	(0.3046)	(0.3889)
News Sentiment Acq. (z)	−0.4724	−1.968***	−1.554*	−9.080***	−0.8905**	−1.161**
	(0.3206)	(0.5284)	(0.9126)	(1.788)	(0.3697)	(0.4376)
N Tweets	−0.0103	−0.0002	−0.0005**	−0.0001**	−0.0003	−0.0002***
	(0.0405)	(0.0002)	(0.0002)	(0.00005)	(0.0004)	(0.00004)
N News Articles	0.1226	0.0119	0.4044	0.0088	0.0031	0.0301
	(0.0853)	(0.0113)	(0.3160)	(0.0110)	(0.0086)	(0.0419)
Mean(LHS)	3.449	3.144	3.453	3.016	3.323	3.793
Coef. Diff. t-Stat. (p-Value)	2.279	(0.023)	0.360	(0.719)	2.295	(0.022)
Observations	2,957	2,322	2,838	2,188	2,979	2,953
R^2	0.1621	0.2755	0.1622	0.2673	0.2472	0.2124
Deal Controls	✓	✓	✓	✓	✓	✓
Firm Controls	✓	✓	✓	✓	✓	✓
Year-by-Quarter FE	✓	✓	✓	✓	✓	✓
Acq. Industry (GIC2) FE	✓	✓	✓	✓	✓	✓

Table VIII

Analyst Conference Calls

This table presents linear probability estimates on the relationship between M&A-related analyst conference calls and the effect of social media reactions on M&A deal withdrawals. The dependent variable is a dummy variable indicating whether an announced M&A transaction was subsequently withdrawn, multiplied by 100. In addition to the same explanatory variables as in Table II, column (1) in Panel A includes the indicator variable *Has Conf. Call (0/1)* which takes the value of one if the acquiring firm held an analyst conference call related to the M&A announcement, and zero otherwise. In column (2) and columns (3) and (4), we split the sample into M&A deals without and with M&A-related analyst conference calls. Column (4) additionally includes measures about the content of the Q&A section of the respective conference call. *% Positive Words*, *% Negative Words*, and *% Constraining Words* are defined as in Loughran and McDonald (2016) and Bodnaruk, Loughran, and McDonald (2015), using the 2022 version of the Loughran and McDonald (2011) Master Dictionary. *Avg. Word Length*, *N Words*, and *Vocabulary* are defined as the average number of letters per word, the number of words, and the number of unique words spoken in the Q&A section of the conference call, respectively. Panel B splits the sample by observations with above- and below-median percentage of *Constrained Words* (columns (1) through (4)), and *Negative Words* (columns (5) through (8)). Panel B additionally distinguishes between words spoken in the presentation (columns (1) and (2), (5) and (6)) and the Q&A section (columns (3) and (4), (7) and (8)) of the conference calls. Each regression includes similar deal and firm-level controls as Table II and year-by-quarter and acquiring-firm industry (GIC two-digit) fixed effects as indicated. *Mean(LHS)* is the average of the dependent variable in the given regression, *Coef. Diff. t-Stat (p-Value)* provides the t-statistic and corresponding p-value testing the hypothesis that the coefficient estimates on our main variable of interest, *Abn. Sentiment (z) (StTw)*, are equal across both regressions. Standard errors are clustered at the year-by-quarter level and are reported in parentheses. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% level, respectively.

Panel A: Analyst Conference Calls and Deal Completion

	𝟙(Deal Withdrawn)			
Sample Split	Full Sample	Conf. Call (0/1)		
	No (1)	Yes (2)	(3)	(4)
Abn. Sentiment (z) (StTw)	−0.7425***	−0.5738**	−1.288**	−1.303**
	(0.2410)	(0.2144)	(0.5848)	(0.5867)
Has Conf. Call (0/1)	−2.563***			
	(0.6202)			
% Positive Words (Q&A)				0.8393
				(1.062)
% Negative Words (Q&A)				−0.1196
				(2.451)

(*Continued*)

Table VIII—*Continued*

Panel A: Analyst Conference Calls and Deal Completion

	𝟙(Deal Withdrawn)			
Sample Split	Full Sample	Conf. Call (0/1)		
	No (1)	Yes (2)	(3)	(4)
% Constraining Words (Q&A)				8.473
				(6.859)
Avg. Word Length (Q&A)				4.339
				(5.457)
N Words (Q&A)				−0.0002
				(0.0017)
Vocabulary (Q&A)				−0.0008
				(0.0127)
Mean(LHS)	3.704	3.612	3.287	3.785
Coef. Diff. t-Stat (p-Value)		1.147	(0.252)	
Observations	3,834	4,928	1,004	1,004
R^2	0.2173	0.2357	0.2059	0.2091
Firm and Deal Controls	✓	✓	✓	✓
Stock Return and Media Controls	✓	✓	✓	✓
Year-by-Quarter FE	✓	✓	✓	✓
Acq. Industry (GIC2) FE	✓	✓	✓	✓

(*Continued*)

Table VIII—*Continued*

Panel B: Conference Calls: Presentation versus Q&A

	𝟙*(Deal Withdrawn)*							
Sample Split	*% Constrained Words*				*% Negative Words*			
	Presentation		Q&A		Presentation		Q&A	
	Low (1)	High (2)	Low (3)	High (4)	Low (5)	High (6)	Low (7)	High (8)
Abn. Sentiment (z) (StTw)	−1.242	−1.447*	0.3123	−2.067**	−1.880	−0.7685	0.4652	−2.310**
	(0.9231)	(0.8030)	(0.5131)	(0.9696)	(1.142)	(0.4897)	(0.4602)	(0.8673)
Mean(LHS)	2.970	2.806	3.006	2.772	3.400	2.381	3.808	1.980
Coef. Diff. t-Stat (p-Value)	0.168	(0.867)	2.169	(0.030)	0.895	(0.371)	2.826	(0.005)
Observations	505	499	499	505	500	504	499	505
R^2	0.2055	0.3400	0.2344	0.3163	0.2637	0.3250	0.2461	0.3325
Deal Controls	✓	✓	✓	✓	✓	✓	✓	✓
Firm Controls	✓	✓	✓	✓	✓	✓	✓	✓
Social & News Media Controls	✓	✓	✓	✓	✓	✓	✓	✓
Stock Return Controls	✓	✓	✓	✓	✓	✓	✓	✓
Conf. Call Controls	✓	✓	✓	✓	✓	✓	✓	✓
Year-by-Quarter FE	✓	✓	✓	✓	✓	✓	✓	✓
Acq. Industry (GIC2) FE	✓	✓	✓	✓	✓	✓	✓	✓

4. The informative component comes from more analysis-rich tweets and fundamental topics (Table VI, Figure 6).

5. Withdrawals that social media "disliked" appear to be value-improving (Table IV).

6.3 Alternative interpretations I would keep in mind

Even with the above, I still consider:

- **Omitted variables:** AbnSent could proxy for privately known deal quality that is also correlated with withdrawal.
- **Managerial attention and selection:** managers may selectively monitor social media more in cases where they already suspect problems.
- **Reverse causality is unlikely short-run, but not impossible:** e.g., if rumors about weak deal commitment circulate quickly and simultaneously affect sentiment.

The paper's design reduces, but cannot fully eliminate, these concerns.

7 Short summary

- A one-standard-deviation increase in abnormal StockTwits sentiment after announcement predicts a **0.64 pp lower** probability of withdrawal (about **16.6%** of the unconditional withdrawal rate).
- The predictive effect is not explained away by announcement CARs, news sentiment/coverage, or analyst recommendation changes.
- The signal strengthens after firms start/activate corporate Twitter accounts, especially for high-follower/verified accounts.
- The informative component comes from fundamental, longer, deal-related content (not memes or technical-trading noise).
- Markets appear to reward the withdrawal of deals that social media initially dislikes.

8 Conclusion

8.1 Summary

This paper concludes that **investor social-media sentiment contains economically meaningful information that is used (directly or indirectly) in corporate decision-making**. In the context of M&A, the authors show that the *initial* social-media reaction to a merger announcement predicts whether the deal is later **withdrawn**, and that the predictive content is not subsumed by standard information channels such as price reactions, traditional media, or analyst recommendations.

The evidence forms a consistent chain:

1. **Social media predicts a real corporate action (deal withdrawal).** Using StockTwits, a one-standard-deviation increase in *abnormal* post-announcement sentiment is associated with an approximately

$$0.64 \text{ percentage point lower probability of withdrawal.}$$

Relative to the average withdrawal rate of about 3.885%, this corresponds to

$$\frac{0.64}{3.885} \approx 16.6\% \quad \text{of the unconditional withdrawal probability.}$$

2. **The result is not explained by conventional signals.** The sentiment–withdrawal relationship remains strong after controlling for announcement abnormal returns (CARs), pre-announcement CARs, traditional-news sentiment and news volume, and analyst recommendation changes, as well as deal and firm controls and fixed effects.

3. **Mechanism evidence is consistent with managerial learning/feedback.** The informativeness of social-media sentiment strengthens after firms begin operating corporate Twitter accounts, and it is particularly strong for firms with more salient/credible engagement (e.g., higher-follower or verified accounts). This pattern is consistent with a lower "cost of listening" increasing the extent to which managers internalize social-media feedback.

4. **The signal is driven by content that looks informative rather than noisy.** The predictive effect is concentrated in longer, acquisition-related posts by fundamental-oriented users, and it is not driven by memes or price-trend/technical-trading content. This reduces the concern that the main result is merely "noise predicting noise".

5. **Withdrawals that social media dislikes look value-improving.** In post-announcement interim returns, the market response is consistent with a "good withdrawal" interpretation: when initial social-media sentiment is more negative, withdrawing (rather than completing) is associated with more favorable abnormal performance, suggesting that social media helps identify deals that are ex ante more likely to destroy value if pursued.

8.2 Why social media can inform corporate decisions

A compact way to understand the paper is that social media provides a fast, low-cost channel for **aggregating dispersed information and criticism** about a proposed corporate action.

1) Social media as an information aggregator. Merger announcements trigger immediate discussion about deal terms, synergies, valuation, integration risk, and regulatory frictions. Even if many individual posts are noisy, the *aggregate* sentiment can reflect a weighted average of:

- investor processing of hard-to-quantify aspects of the deal (fit, complexity, strategic logic),
- incremental information from specialized investors (industry knowledge, regulatory knowledge),
- and rapid peer-to-peer scrutiny of managerial narratives.

2) A simple learning model for withdrawal. Let the manager decide whether to proceed (Proceed = 1) or withdraw (Proceed = 0). Suppose the manager has a prior mean value μ_0 with precision τ_0 and observes two noisy signals: a market-price-based signal s_P with precision τ_P and a social-media signal s_S with precision τ_S. A stylized posterior mean is

$$\mu_1 = \frac{\tau_0\mu_0 + \tau_P s_P + \tau_S s_S}{\tau_0 + \tau_P + \tau_S}.$$

If the manager withdraws when μ_1 falls below a cutoff (reflecting reputation, break fees, and opportunity costs), then **more negative social-media sentiment** (lower s_S) raises the withdrawal probability. The paper's baseline regressions estimate this reduced-form implication.

3) Why the effect strengthens after corporate Twitter adoption. Corporate Twitter adoption increases exposure to and monitoring of social discourse, effectively raising the precision (or the weight) of s_S from the manager's perspective. In the simple posterior above, this is an increase in τ_S, which makes the decision more sensitive to social-media sentiment:

$$\tau_S \uparrow \Rightarrow \frac{\partial \mu_1}{\partial s_S} \uparrow \Rightarrow \text{ sentiment has a larger effect on withdrawal decisions.}$$

4) Why "fundamental" and longer posts matter. Longer acquisition-related posts are more likely to contain structured reasoning (deal terms, integration concerns, valuation arguments), so they plausibly have higher signal-to-noise. In the learning framework, these messages raise the effective precision of s_S. In contrast, memes and price-trend chatter are low-precision signals and should not strongly affect rational decision-making.

5) Why social media is not redundant with announcement CAR. Announcement CAR mixes multiple objects: expected synergies, financing implications, completion probability, and market-wide discount-rate movements. Social-media discourse can reveal *deal-quality concerns* that are not immediately or fully impounded into prices, or that are impounded but partially offset by other price-moving forces. Hence sentiment can retain incremental predictive power even conditional on CAR.

8.3 Takeaway

Cookson, Niessner, Schiller (JF, 2026) show that abnormal investor social-media sentiment around merger announcements robustly predicts later deal withdrawals (a 1 SD sentiment increase lowers withdrawal probability by about 0.64 pp, roughly 16.6% of the mean), the effect strengthens when managers have better access to social media via corporate Twitter accounts, and the signal is driven by fundamentally oriented acquisition-related content, consistent with social media providing actionable feedback that helps firms abandon value-destroying deals.